

20.1.2 Matching Currents in the Mirror

Many analog applications are susceptible to errors due to layout. In circuits in which devices need to be matched, layout becomes a critical factor. For example, in the basic current mirror shown in Fig. 20.2, first-order process errors can cause the output current, I_O , to be significantly different from the reference current. Process parameters such as gate-oxide thickness, lateral diffusion, oxide encroachment, and oxide charge density can drastically affect the performance of a device. Layout methods can be used to minimize the first-order effects of these parameter variations.

Threshold Voltage Mismatch

In a given current mirror application, the values for the threshold voltages are critical in determining the overall accuracy of the mirror. Again, examine the basic current mirror shown in Fig. 20.2. Since both devices have the same value for V_{GS} and assuming the sizes and transconductance parameters for both devices are equal, let's examine the effect of a mismatch in threshold voltages between the two devices. If it is assumed that the threshold mismatch is distributed across both devices such that

$$V_{THN1} = V_{THN} - \frac{\Delta V_{THN}}{2} \quad (20.5)$$

$$V_{THN2} = V_{THN} + \frac{\Delta V_{THN}}{2} \quad (20.6)$$

where V_{THN} is the average value of V_{THN1} and V_{THN2} and ΔV_{THN} is the mismatch, then

$$\frac{I_O}{I_{REF}} = \frac{\frac{KP_n W}{2 L} (V_{GS} - V_{THN} - \frac{\Delta V_{THN}}{2})^2}{\frac{KP_n W}{2 L} (V_{GS} - V_{THN} + \frac{\Delta V_{THN}}{2})^2} = \frac{\left[1 - \frac{\Delta V_{THN}}{2(V_{GS} - V_{THN})}\right]^2}{\left[1 + \frac{\Delta V_{THN}}{2(V_{GS} - V_{THN})}\right]^2} \quad (20.7)$$

If both expressions are squared and the higher order terms are ignored, then the first-order expression for the ratio of currents becomes

$$\frac{I_O}{I_{REF}} \approx 1 - \frac{2\Delta V_{THN}}{V_{GS} - V_{THN}} = 1 - \frac{2\Delta V_{THN}}{V_{DS,sat}} \quad (20.8)$$

Equation (20.8) is quite revealing because it shows that as V_{GS} decreases, the difference in the mirrored currents increases due to threshold voltage mismatch. This is particularly critical for devices that are separated by relatively long distances because the threshold voltage is susceptible to process gradients. *To attain high speed and to reduce the effects of threshold voltage mismatch, a large gate overdrive voltage should be used* (remembering for a long-channel process that $V_{ovn} = V_{DS,sat} = V_{GS} - V_{THN}$). Of course the drawback, for a current mirror, is a reduced range of compliance (the MOSFET enters the triode region earlier).

Transconductance Parameter Mismatch

The same analysis can be performed on the transconductance parameter, KP_n . If $KP_{n1} = KP_n - \Delta KP_n/2$ and $KP_{n2} = KP_n + \Delta KP_n/2$, where KP_n is the average of KP_{n1} and KP_{n2} , then assuming perfect matching on all other parameters, the difference in the currents becomes

$$\frac{I_O}{I_{REF}} = \frac{KP_n + 0.5\Delta KP_n}{KP_n - 0.5\Delta KP_n} \approx 1 + \frac{\Delta KP_n}{KP_n} \quad (20.9)$$

Since KP_n is a process parameter, we might think that as CMOS scales downwards (C'_{ox} goes up) the mismatch due to oxide variations and mobility differences might get better. However, there is less averaging of these variations with the smaller layout sizes. Practically, as we're about to see, differences in V_{DS} dominate the matching behavior.

Drain-to-Source Voltage and Lambda

One aspect of current mirror design that is *critical* for generating accurate currents is the drain-to-source voltage. As seen in Fig. 20.4, the only point where the currents of the devices are actually the same is when their V_{DS} values are equal. In Eq. (20.3) the ratio of the output current to the reference current is affected by both the matching in the drain-to-source voltages (V_O and V_{DS1}) and the device λ s. If, for example in the short-channel process (see Table 9.2), $V_{DS1} = .35$ V, $V_{DS2} = V_O = .75$ V, and $\lambda_1 = \lambda_2 = 0.6$ V⁻¹, then

$$\frac{I_O}{I_{REF}} = \frac{1 + \lambda_2 \cdot V_O}{1 + \lambda_1 \cdot V_{DS1}} = \frac{1 + 0.6 \cdot 0.75}{1 + 0.6 \cdot 0.35} = 1.20 \quad (20.10)$$

resulting in 20% error! It is extremely important, for good matching, that the V_{DS} values of the MOSFETs in the current mirror are equal.

Layout Techniques to Improve Matching

Most general analog applications require the length of the gate to be longer than minimum since the channel-length modulation, λ , has less effect on longer devices than on shorter ones (as discussed in Ch. 9). As a result, the minimum-sized devices found in digital circuits are not used as often in general analog design (however, see Ch. 26). However, the larger devices can result in larger parasitics if some layout issues are not considered. Figure 20.5a illustrates a basic MOSFET device with a large W/L . The implant resistance of the source and drain can be modeled as shown in Fig. 20.5b. The implant resistance can easily be reduced by simply adding as many contacts as possible along the width of both the source and the drain as seen in Fig. 20.5c. The increase in the number of contacts results in lower resistance, more current capability, and a more distributed current load throughout the device. However, as the device width increases, another technique is used which distributes the parasitics (both resistive and capacitive) into smaller contributions.

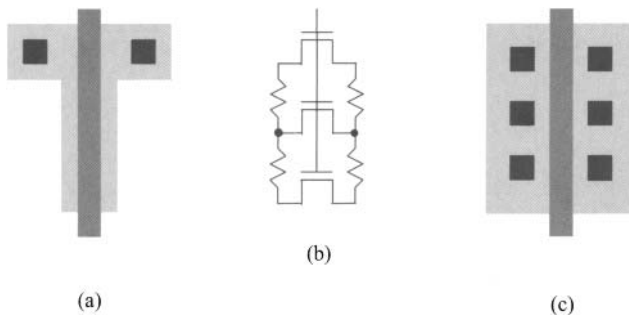


Figure 20.5 (a) Large device with a single contact and (b) its equivalent circuit. (c) Adding more contacts to reduce parasitic resistance.

Examine Fig. 20.6. In this figure, a single device with a large W/L is split into several parallel devices, each with a width one-fourth of the original W . One result of splitting the device into several parts is smaller overall parasitic capacitance associated with the reverse-biased implant substrate diode (the drain or source depletion capacitance to substrate). Since values of C_{db} and C_{sb} are proportional to W , the split device reduces these parasitics by a factor of $(n+1)/2n$ where n is the number of parallel devices and is odd. If n is even, then the C_{sb} is reduced by one-half and C_{db} is reduced by $(n+2)/2n$. As seen in Fig. 20.6b, putting the devices in parallel also reduces the parasitic resistance in series with both the source and the drain (it gets cut roughly in half too).

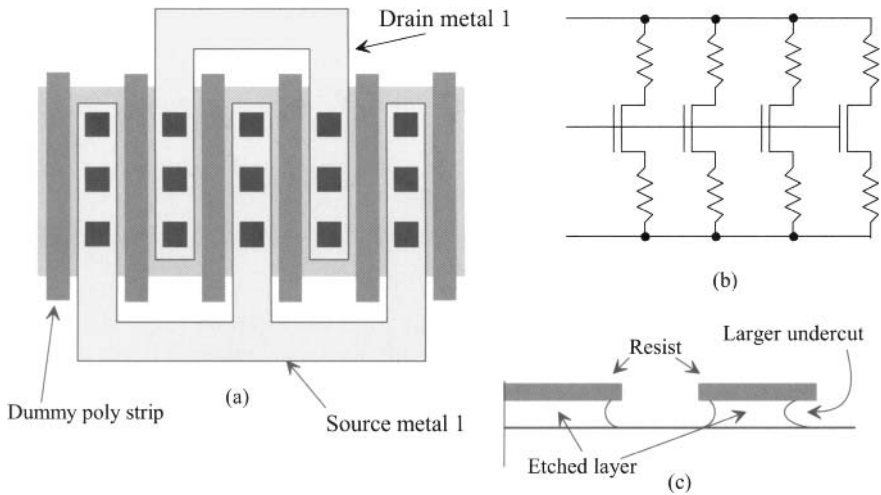


Figure 20.6 (a) A parallel device with dummy strips, (b) the equivalent circuit, and (c) undercutting.

Notice also, that Fig. 20.6a has dummy poly strips on both sides of the device. These strips are used to help minimize the effects of undercutting the poly on the outer edges after patterning, Fig. 20.6c. If the dummy strips had not been used, the poly would have been etched out more under the outermost gates, resulting in a mismatch between the four parallel devices.

When matching two devices, it is imperative that the two devices be as symmetrical as possible. Always orient the two devices in the same direction, unlike those illustrated in Fig. 20.7.

Splitting the devices into parallel devices and interdigitizing them can distribute process gradients across both devices and thus improve matching. An example of this is seen in Fig. 20.8. In (a) the current mirror seen in (b) is laid out. Each MOSFET in (b) is split up into four MOSFETs. If the W/L of each MOSFET in (b) is $80/2$, then the size of each MOSFET (finger) in (a) is $20/2$. Note the use of dummy poly strips on this layout. A good exercise at this point is to lay out the mirror of Fig. 20.8 in a common-centroid arrangement (see Ch. 5). Using a large layout area (long lengths and wide widths) and common-centroid layouts can result in significant improvements in matching.

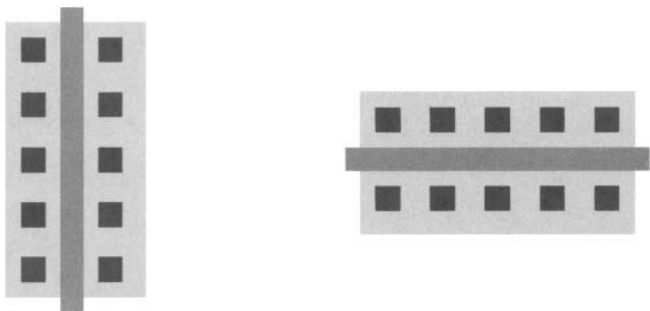


Figure 20.7 Devices with differing orientation (bad).

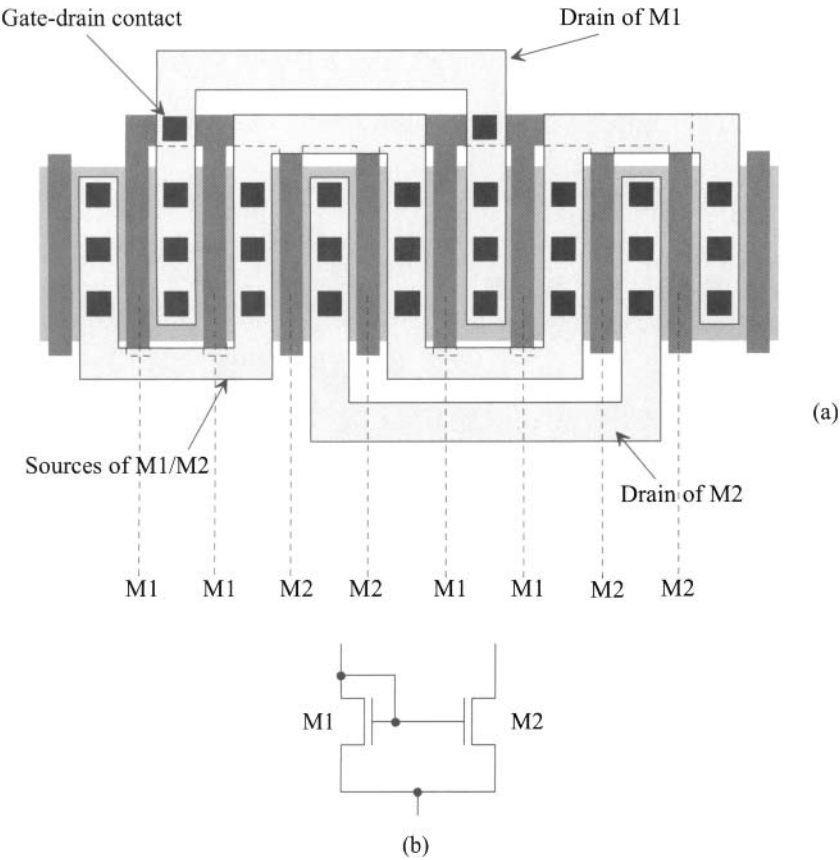


Figure 20.8 (a) Layout of a simple current mirror using interdigitation and (b) equivalent circuit.

Layout of the Mirror with Different Widths

When laying out the current mirror, the lateral diffusion, L_{diff} , under the gate oxide (Fig. 5.13) and the oxide encroachment, W_{enc} , can affect the actual length and width, respectively, of the MOSFET. If not careful with the layout, this will affect the mirror's ratio. Equation 20.3 can be rewritten, without including the differences due to drain-source voltage and lambda, as

$$\frac{I_O}{I_{REF}} = \frac{(W_{2drawn} - 2W_{enc}) \cdot (L_{1drawn} - 2L_{diff})}{(W_{1drawn} - 2W_{enc}) \cdot (L_{2drawn} - 2L_{diff})} \quad (20.11)$$

If the requirement that $L_{1drawn} = L_{2drawn}$ is imposed, then the widths of the devices determine the relative currents in the mirror. Figure 20.9a shows a current mirror layout without width compensation. If W_{enc} is 0.1, then for this layout,

$$\frac{I_O}{I_{REF}} = \frac{40 - .2}{20 - .2} = 2.01 \text{ (a 1\% error due to poor layout)}$$

Figure 20.9b shows how to lay out a current mirror to avoid these problems. The layout of M2 is two MOSFETs in parallel. This can be specified in SPICE by adding $M = X$ after the MOSFET statement in the netlist, where X is the number of MOSFETs,

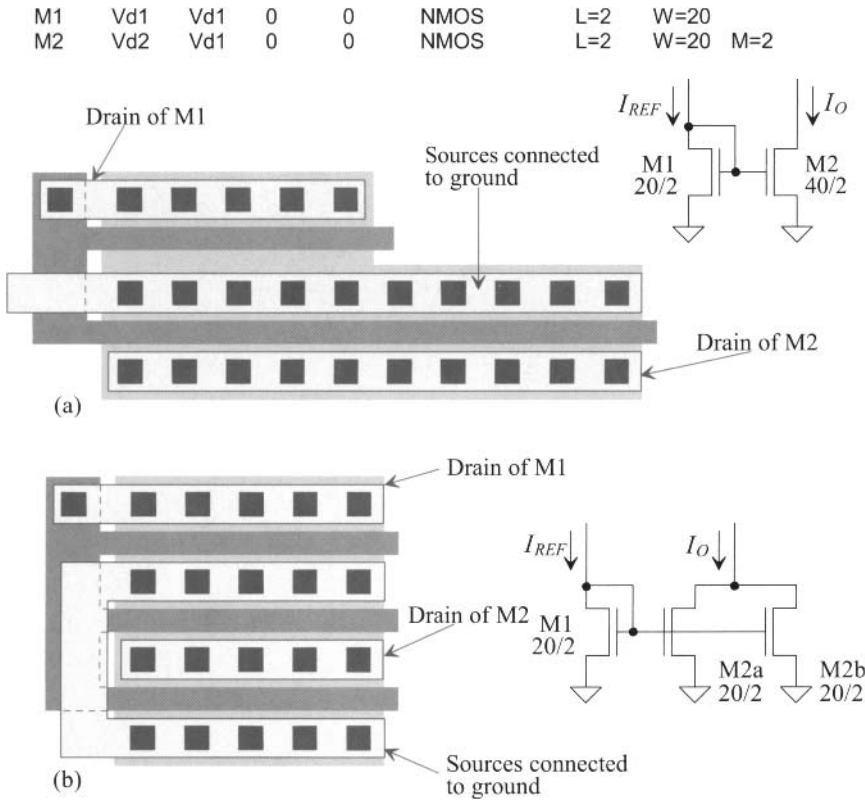


Figure 20.9 Layout of a current mirror (a) without width correction and (b) with width correction.

20.1.3 Biasing the Current Mirror

Using a resistor to set the bias current, as seen in Figs. 20.2–20.4, can result in currents that are too dependent on the power supply value and temperature. Consider the current mirror seen in Fig. 20.10a. In this design we’ve used the sizes and bias current given in Table 9.2 (the short-channel CMOS process) to select the resistor. In particular, $V_{SG} = 0.35\text{ V}$ so the gate potential is 0.65 V . Figure 20.10b shows how the reference and output currents vary if V_{DD} is swept from 900 mV to 1 V. The reference current is linearly dependent on V_{DD} (as seen in Ex. 20.1). The output current is dependent on both the reference current and the V_{DS} (λ) of M2. In general, we want a reference current that isn’t dependent on power supply or ground variations (noise). This is an important point. Consider the following example.

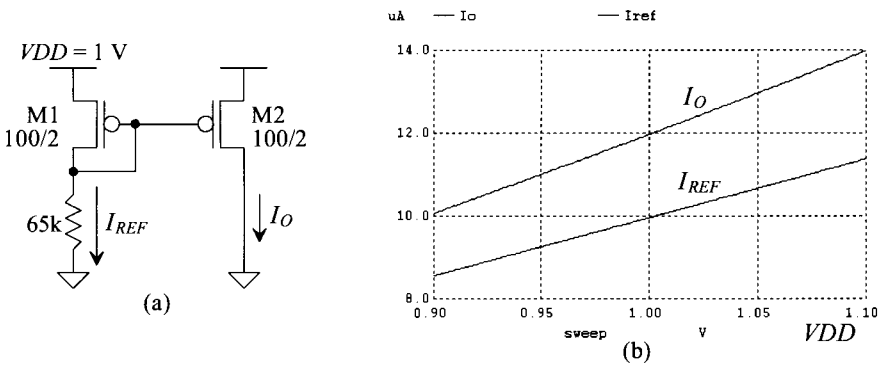


Figure 20.10 How reference and output current vary with V_{DD} .

Example 20.2

Regenerate Fig. 20.10 if the 65k resistor is replaced with an ideal $10\text{ }\mu\text{A}$ current source, Fig. 20.11a. Explain the results.

The simulation results are seen in Fig. 20.11b. Note how the output current variation with V_{DD} is much better than what is seen in Fig. 20.10. The reference current through M1 is, of course, a constant. The output current is larger than the

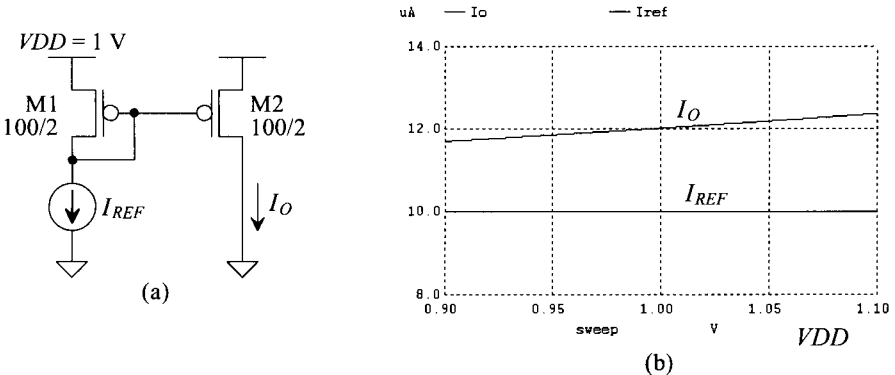


Figure 20.11 Showing that V_{DD} variations don’t affect the reference current.

reference current because $V_{SD1} < V_{SD2}$ (notice how often this point is coming up in the design of current mirrors). The reason I_o varies is due to the finite output resistance ($\lambda \neq 0$) of M2, Eq. (20.2).

The point of this example is that it is desirable to have a bias current that doesn't vary with changes in V_{DD} or ground. ■

Using a MOSFET-Only Reference Circuit

Let's try replacing the resistor bias with a MOSFET, Fig. 20.12. If we assume long-channel behavior, then

$$V_{DD} = V_{SG3} + V_{GS1} \quad (20.12)$$

or

$$V_{DD} = \sqrt{\frac{2I_{REF}}{KP_p \frac{W_3}{L_3}}} + V_{THP} + \sqrt{\frac{2I_{REF}}{KP_n \frac{W_1}{L_1}}} + V_{THN} \quad (20.13)$$

If we are going to think of M3 as a resistor, as in Fig. 20.2, so that we mirror the current in M1, then we solve for the size of M3. Using the data from Table 9.1, we get

$$5 = \sqrt{\frac{2 \cdot 20}{40 \cdot \frac{W_3}{L_3}}} + 0.9 + \sqrt{\frac{2 \cdot 20}{120 \cdot \frac{10}{2}}} + 0.8 \rightarrow \frac{W_3}{L_3} = 0.11 \approx \frac{10}{90} \quad (20.14)$$

If we are going to think of M1 as a resistor, as in Fig. 20.10a, so that we mirror the current in M3, then we solve for the size of M1 as we did in Eq. (20.14) and get

$$\frac{W_1}{L_1} \approx \frac{10}{270} \quad (20.15)$$

Figure 20.13 shows the operation of these MOSFET bias circuits with V_{DD} swept from 4.5 to 5.5 V. Note that, at 5 V, we should have a bias current of 20 μA . What we get is 14 μA . The reason that we have such a large difference is due to neglecting the output resistance of the MOSFETs (and the effects of mobility degradation). Note that the sensitivities, how the output current changes with V_{DD} , are roughly 8 $\mu\text{A}/\text{V}$. (Let's compare this to hand-calculated values of sensitivity next.)

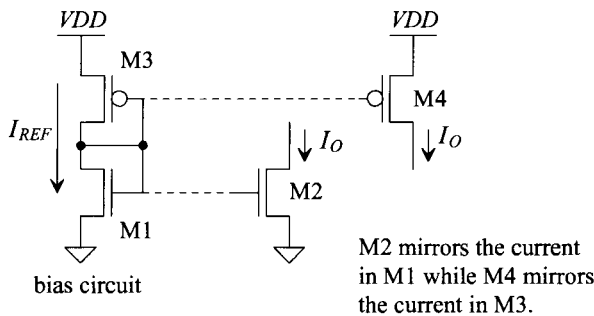


Figure 20.12 A MOSFET-only bias circuit.

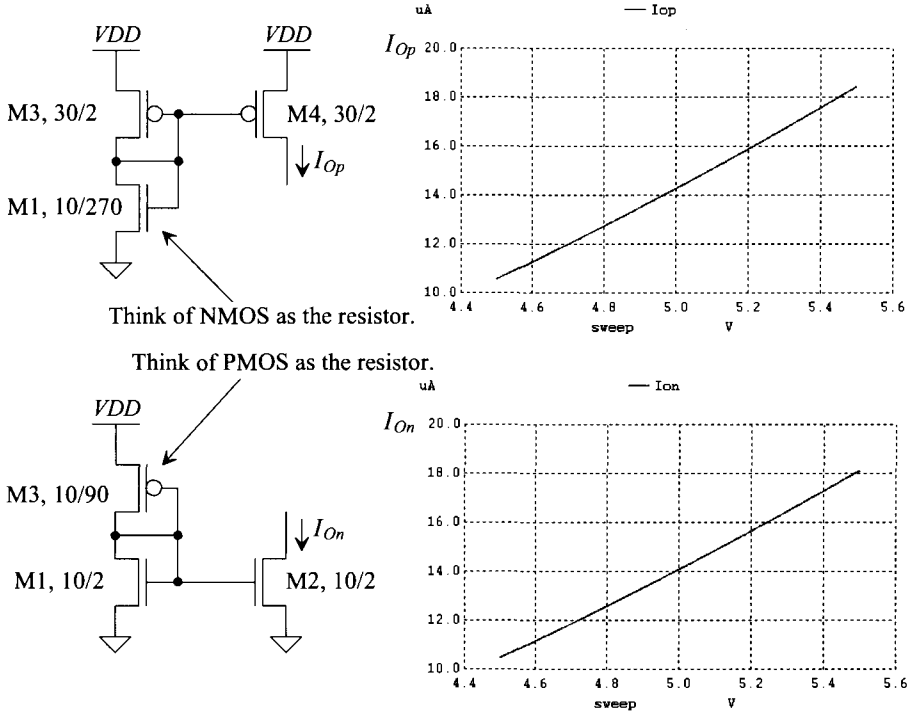


Figure 20.13 Behavior of MOSFET-only bias circuits with changes in V_{DD} .

To determine I_{REF} 's sensitivity to V_{DD} , let's take the derivative of the I_{REF} in Eq. (20.13) with respect to V_{DD} . First let's write

$$(V_{DD} - V_{THN} - V_{THP})^2 = I_{REF} \cdot \left[\sqrt{\frac{2L_3}{K P_p \cdot W_3}} + \sqrt{\frac{2L_1}{K P_n \cdot W_1}} \right]^2 \quad (20.16)$$

and next

$$\frac{\partial I_{REF}}{\partial V_{DD}} = \frac{2 \cdot V_{DD}}{K} - \frac{2 \cdot (V_{THN} + V_{THP})}{K} \quad (20.17)$$

Using the values in Table 9.1 and Eqs. (20.14) or (20.15), K is approximately $547 \times 10^3 \text{ V}^2/\text{A}$ so

$$\frac{\partial I_{REF}}{\partial V_{DD}} = 12 \text{ } \mu\text{A/V} \quad (20.18)$$

For every millivolt change in V_{DD} (around $V_{DD} = 5 \text{ V}$), we get 12 nA change in I_{REF} . As a comparison from Ex. 20.1, we can write (assuming the change in V_{GS} with V_{DD} small)

$$\frac{\partial I_{REF}}{\partial V_{DD}} \approx \frac{1}{R} = \frac{1}{200k} = 5 \text{ } \mu\text{A/V} \quad (20.19)$$

For every millivolt change in V_{DD} we get 5 nA change in bias current.

Supply Independent Biasing

Instead of putting the resistor in the drain side of the current mirror, let's consider placing it in the source side, Fig. 20.14a. One of the problems with this approach is that if we try to mirror the current in M2 using M5 we won't know the value of the mirrored current. It's not obvious how V_{GS2} and V_{GS5} are related. In (b) we add a diode connected M1 so that its current can be mirrored (by M5). The next question is how do we force the same current through both M1 and M2? Figure 20.14c shows the addition of a PMOS current mirror to do this. From (c) we can write

$$V_{GS1} = V_{GS2} + I_{REF} \cdot R \quad (20.20)$$

which can only be valid if $V_{GS1} > V_{GS2}$. To ensure that this is the case, we use a larger value of β in M2, that is, we *multiply-up* M1's β in M2 so that less gate-source voltage is needed to conduct I_{REF} . Generally, this is done by simply using a larger width in M2. The resulting circuit is called a *Beta-multiplier* reference circuit. Knowing

$$V_{GS} = \sqrt{\frac{2I_D}{\beta}} + V_{THN} \quad (20.21)$$

$$\left(\beta = KP_n \cdot \frac{W}{L} \right) \text{ and}$$

$$\beta_2 = K \cdot \beta_1 \text{ (which is satisfied by } W_2 = K \cdot W_1 \text{)} \quad (20.22)$$

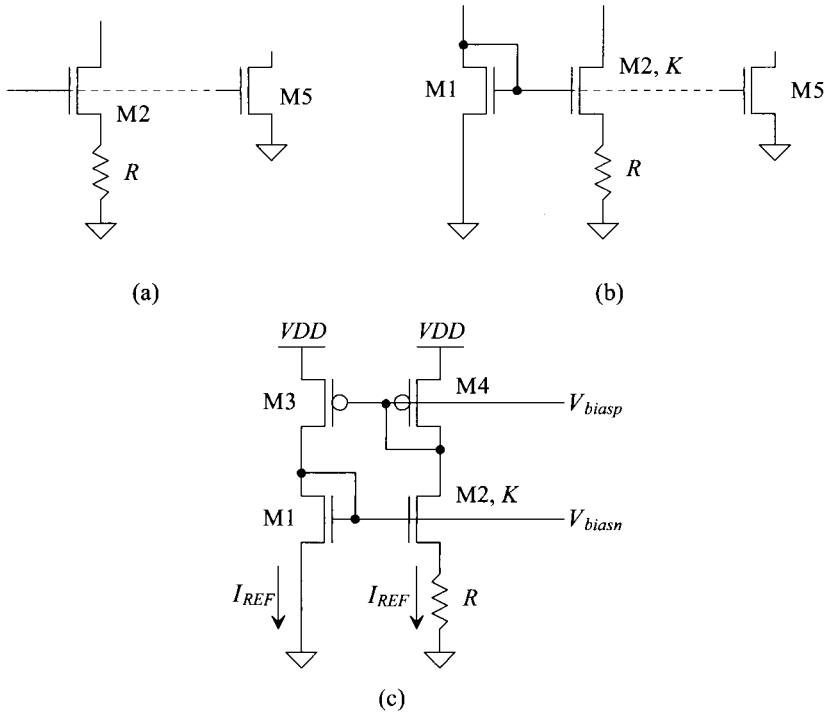


Figure 20.14 Developing the Beta-multiplier reference.

we can write

$$I_{REF} = \frac{2}{R^2 K P_n \cdot \frac{W_1}{L_1}} \left(1 - \frac{1}{\sqrt{K}}\right)^2 \quad \text{or} \quad V_{DS,sat} = V_{GS} - V_{THN} = \frac{2}{R \cdot K P_n \cdot \frac{W_1}{L_1}} \left(1 - \frac{1}{\sqrt{K}}\right) \quad (20.23)$$

It's important to note I_{REF} and V_{ovn} ($= V_{GS} - V_{THN}$), neglecting the finite output resistance of the MOSFETs which wasn't included in the derivation of Eq. (20.23), are independent of V_{DD} . Solving for R with the values given in Table 9.1 and a K of 4 gives $R = 6.5 \text{ k}\Omega$. Note that this circuit, when $K = 4$, is sometimes called a *constant- g_m* bias circuit because

$$g_m = \sqrt{2 K P_n \frac{W}{L} \cdot I_{REF}} = \frac{1}{R} \quad (20.24)$$

a constant independent of MOSFET process shifts. Figure 20.15 shows the schematic of a bias circuit based on the sizes, bias currents, and $V_{DS,sat}$ given in Table 9.1.

Note that a “start-up circuit” was included in Fig. 20.15. In any self-biased circuit there are two possible operating points: the one we just described and the unwanted one where zero current flows in the circuit. This unwanted state occurs when the gates of M1/M2 are at ground while the gates of M3/M4 are at V_{DD} . When in this state, the gate of MSU1 is at ground and so it is off. The gate of MSU2 is somewhere between V_{DD} and $V_{DD} - V_{THP}$. MSU3, which behaves like an NMOS switch, turns on and leaks current into the gates of M1/M2 from the gates of M3/M4. This causes the current to snap to the desired state and MSU3 to turn off. *Note that during normal operation the start-up circuit should not affect the Beta-multiplier's operation.* The current through MSU3 should be zero (or very small).

The Beta-multiplier is an example of a circuit that uses positive feedback. The addition of the resistor kills the closed loop gain (a positive feedback system can be stable if its closed loop gain is less than one). However, if we decrease the size of the resistor, we increase the gain of the loop and push the feedback system closer to instability. An example of when this could occur is if the parasitic capacitance on the source of M2 to ground is large (effectively shorting M2's source to ground). If the resistor, for example, is bonded out off-chip to set the current, it is likely that this bias circuit will oscillate.

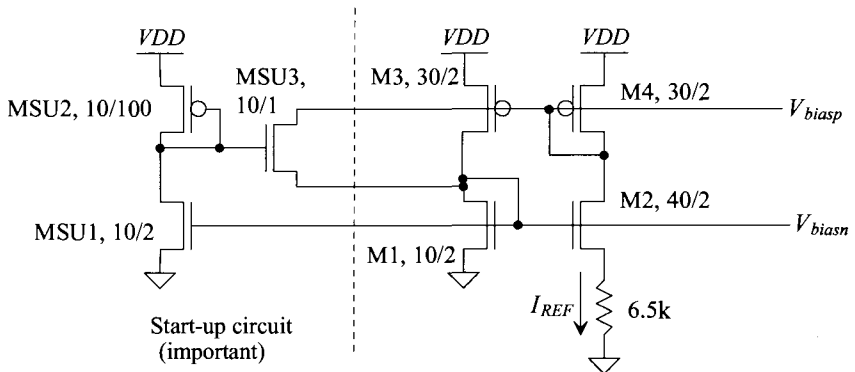


Figure 20.15 Beta-multiplier reference for biasing in the long-channel process described in Table 9.1.

Figure 20.16 shows how the reference currents through M1 and M2 vary with V_{DD} . The minimum value of V_{DD} can be estimated by looking at the minimum value of voltage across the drain-source of M3 and M1. For M3 this is $V_{SD3,sat}$ or 250 mV (because we are using the parameters from Table 9.1 in the design of this current reference). For M1 this is $V_{DS1} = V_{GS1} = 1.05$ V. We can then write

$$V_{DD\min} = V_{SD3,sat} + V_{GS1} = 1.3 \text{ V} \quad (20.25)$$

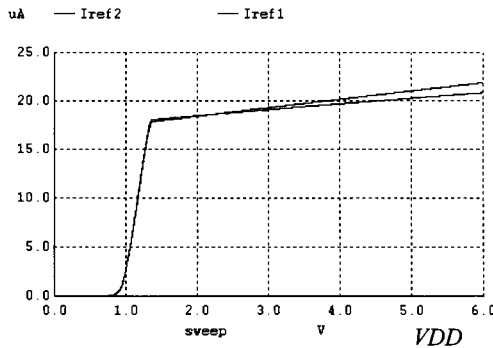


Figure 20.16 The reference currents through M1 and M2 in the Beta-multiplier.

Finally, the sensitivity of I_{REF} is directly dependent on the output resistance of the MOSFETs. From the simulations

$$\frac{\partial I_{REF}}{\partial V_{DD}} \approx 800 \text{ nA/V}$$

or almost an order of magnitude better than the previous reference circuits

Example 20.3

Estimate the voltage on the gate of M6 and its drain current in Fig. 20.17.

M3 and M4 are biased to source 20 μA of current. Since $V_{GS1} = V_{GS2} = V_{DS1}$ (and M1/M2 have the same drain current, see Fig. 20.1 and the associated discussion), it follows that $V_{GS6} = V_{DS2} = V_{GS1} = 1.05$ V. We treat M6, for biasing purposes, as if its gate were tied to the gates of M1 or M2. It follows then that $I_{D6} = 20 \mu\text{A}$. ■

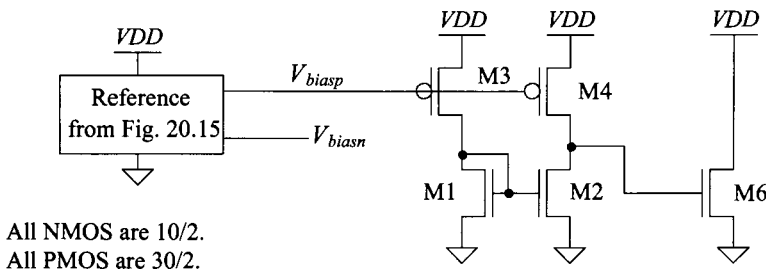


Figure 20.17 Circuit used in Ex. 20.3